



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2026
SECONDARY 4 EXPRESS/ 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

4049/02

Paper 2

2 hours 15 minutes

Marking Scheme

Question	Answer	Marks	Partial Marks	Guidance
1(a)	$\cos 75^\circ = \cos(45^\circ + 30^\circ)$ $= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$ $= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$ $= \frac{\sqrt{6} - \sqrt{2}}{4}$	4	M1 B1, B1 AG1	Addition formulae Any two correct trigo ratios in surd form
1(b)	$\sin^2 75^\circ + \cos^2 75^\circ = 1$ $\sin^2 75^\circ = 1 - \cos^2 75^\circ$ $= 1 - \left(\frac{\sqrt{6} - \sqrt{2}}{4}\right)^2$ $= 1 - \frac{6 + 2 - 2\sqrt{12}}{16}$ $= 1 - \frac{8 - 4\sqrt{3}}{16}$ $= \frac{8 + 4\sqrt{3}}{16}$ $= \frac{2 + \sqrt{3}}{4}$	3	M1 M1 A1	Use of pythagorean identities Expansion

Question	Answer	Marks	Partial Marks	Guidance
2	$kx^2 + (k-1)x + 4 = 2x + 3$ $kx^2 + (k-3)x + 1 = 0$ $(k-3)^2 - 4k < 0$ $k^2 - 6k + 9 - 4k < 0$ $k^2 - 10k + 9 < 0$ $(k-9)(k-1) < 0$ $1 < k < 9$ Since $k < 0$, there is no real value for which the curve lies completely below the line.	5	B1 M1 M1 A1 AG1	Equating Discriminant Solving quadratic inequalities With condition $k < 0$

Question	Answer	Marks	Partial Marks	Guidance
3	$2 \tan^2 x + \sec x - 4 = 0$ $2(\sec^2 x - 1) + \sec x - 4 = 0$ $2 \sec^2 x + \sec x - 6 = 0$ $(2 \sec x - 3)(\sec x + 2) = 0$ $\sec x = \frac{3}{2}$ or $\sec x = -2$ $\cos x = \frac{2}{3}$ or $\cos x = -\frac{1}{2}$ $x = 0.84107$ or $\pi - 0.84107$ or $x = \pi - \frac{\pi}{3}$ or $\pi + \frac{\pi}{3}$ $= 0.841$ or 2.30 (3 s.f.) or $= \frac{2\pi}{3}$ or $\frac{4\pi}{3}$	7	B1 M1 A1 B1 M1 A1, A1	Pythagorean identities Solving quadratic equation $\sec x = \frac{1}{\cos x}$ Finding basic angle for either For each correct pair/ need not be in exact form

Question	Answer	Marks	Partial Marks	Guidance												
4(a)	<table border="1"><tr><td>t</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$\ln N$</td><td>7.09</td><td>8.19</td><td>9.29</td><td>10.39</td><td>11.49</td></tr></table>	t	1	2	3	4	5	$\ln N$	7.09	8.19	9.29	10.39	11.49	3	B1	Appropriate scale – ¾ of the grid covered
	t	1	2	3	4	5										
	$\ln N$	7.09	8.19	9.29	10.39	11.49										
	$N = ab^t$ $\ln N = (\ln b)t + \ln a$	B1	Points plotted accurately													
	B1	Best-fit line														
4(b)	Intercept: $\ln N = 6$ $\ln a = 6$ $a = e^6 = 403$ (3 s.f.)	2	M1 A1	Student’s vertical intercept												
4(c)	$\text{gradient} = \frac{11.5 - 7}{5 - 1} = 1.125$ $\ln b = 1.125$ $b = e^{1.125} = 3.08$ (3 s.f.)	2	M1 A1	Gradient based on graph												

Question	Answer	Marks	Partial Marks	Guidance
5(a)	$E = -0.01\theta^2 + 0.4\theta + 2$ $= -0.01(\theta^2 - 40\theta) + 2$ $= -0.01[(\theta - 20)^2 - 20^2] + 2$ $= -0.01(\theta - 20)^2 + 6$ $= 6 - 0.01(\theta - 20)^2$	3	B1, B1, B1	$a = 6, b = -0.01, c = 20$
5(b)	Maximum output = 6 kWh when $\theta = 20^\circ$.	2	$\sqrt{B1}$, $\sqrt{B1}$	From students' (a)
5(c)	Since E is a quadratic function, halving the angle will not halve the energy output. OR $E = 6 - 0.01(10 - 20)^2 = 5\text{kWh} \neq 3\text{kWh}$. Therefore, the claim is incorrect.	1	B1	Any reasonable explanation that indicates non-linear relationship

Question	Answer	Marks	Partial Marks	Guidance
6(a)	$f(x) = x^3 + px^2 + qx + 6$ $f(2) = 0$ $2^3 + p(2)^2 + 2q + 6 = 0$ $4p + 2q = -14$ $2p + q = -7 \text{ --- (1)}$ $f(-1) = 18$ $(-1)^3 + p(-1)^2 - q + 6 = 18$ $p - q = 13 \text{ --- (2)}$ (1) + (2): $3p = 6$ $p = 2$ $q = -11$	5	M1 M1 A1 M1 A1	Factor theorem Remainder theorem For either equation in equivalent form Simultaneous equations
6(b)	$P(x) = 2x^3 - 5x^2 + x + 3$ $= (2x - 3)(x^2 - x - 1)$ $P(x) = 0$ $(2x - 3)(x^2 - x - 1) = 0$ $2x - 3 = 0 \text{ or } x^2 - x - 1 = 0$ $x = \frac{3}{2}$ $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$ $= \frac{1 \pm \sqrt{5}}{2}$ $x = \frac{3}{2}, \frac{1 + \sqrt{5}}{2} \text{ or } \frac{1 - \sqrt{5}}{2}$	4	M1 M1 A1, A1	Finding quadratic factor, either long division or comparing coefficient Quadratic formula / completing the square for $\frac{3}{2}$ and for the other pair

Question	Answer	Marks	Partial Marks	Guidance
7(a)	$\sin \theta = \frac{BD}{10}$ $\cos \theta = \frac{BE}{3}$ $BD = 10 \sin \theta$ $BE = 3 \cos \theta$ Height, $h = 10 \sin \theta - 3 \cos \theta + 1.5$	3	M1, M1 AG1	ratio from each triangle
7(b)	$10 \sin \theta - 3 \cos \theta = R \sin(\theta - \alpha)$ $R = \sqrt{10^2 + 3^2}$ $= \sqrt{109}$ $\alpha = \tan^{-1}\left(\frac{3}{10}\right)$ $= 16.699^\circ$ (3 d.p.) $10 \sin \theta - 3 \cos \theta + 1.5 = 1.5 + \sqrt{109} \sin(\theta - 16.7^\circ)$	4	M1 A1 M1 A1	Finding R Finding α
7(c)	$1.5 + \sqrt{109} \sin(\theta - 16.699^\circ) = 5$ $\sqrt{109} \sin(\theta - 16.699^\circ) = 3.5$ $\sin(\theta - 16.699^\circ) = \frac{3.5}{\sqrt{109}}$ $\theta - 16.699^\circ = 19.587^\circ$ $\theta = 36.3^\circ$ (1 d.p.)	2	M1 A1	Simplified trigo equation

Question	Answer	Marks	Partial Marks	Guidance
8	$\frac{dy}{dx} = \frac{a}{x} + bx - \frac{4}{x^2}$ When $x = 1$, $\frac{dy}{dx} = 0$. $a + b - 4 = 0 \Rightarrow a + b = 4 \text{ --- (1)}$ $\frac{d^2y}{dx^2} = -\frac{a}{x^2} + b + \frac{8}{x^3}$ When $x = 2$, $\frac{d^2y}{dx^2} = 0$ $-\frac{a}{4} + b + \frac{8}{8} = 0 \Rightarrow 4b - a = -4 \text{ --- (2)}$ (1) + (2): $5b = 0 \Rightarrow b = 0$ $a = 4$ $\frac{dy}{dx} = \frac{4}{x} - \frac{4}{x^2}$ $y = \int \frac{4}{x} - \frac{4}{x^2} dx$ $= 4 \ln x + \frac{4}{x} + c$ $6 = 4 \ln 1 + \frac{4}{1} + c$ $c = 4 + c$ $c = 2$ $\therefore y = 4 \ln x + \frac{4}{x} + 2$	9	M1 M1 M1 M1 A1 M1, M1 M1 A1	Sub. $x = 1$ into first derivative = 0 Get 2nd derivative Uses $x = 2$ on 2nd derivative = 0 Solve simultaneously 2 equations $a = 4$ and $b = 0$ For ln function, and reciprocal function Uses (1,6) in y Or equivalent

Question	Answer	Marks	Partial Marks	Guidance
10(a)	$\text{distance} = \sqrt{x^2 + y^2}$ $= \sqrt{x^2 + (2x-1)^2}$ $= \sqrt{x^2 + 4x^2 - 4x + 1}$ $= \sqrt{5x^2 - 4x + 1}$	2	M1 AG1	Subst. $y = 2x - 1$ into distance Simplifies into expression
10(b)	$r = \sqrt{5x^2 - 4x + 1} = (5x^2 - 4x + 1)^{\frac{1}{2}}$ $\frac{dr}{dx} = \frac{1}{2}(5x^2 - 4x + 1)^{-\frac{1}{2}}(10x - 4)$ $= \frac{5x - 2}{\sqrt{(5x^2 - 4x + 1)}}$ $\frac{dx}{dt} = 0.05$ $\frac{dr}{dt} = \frac{dr}{dx} \times \frac{dx}{dt}$ $= \frac{5x - 2}{\sqrt{(5x^2 - 4x + 1)}} \times 0.05$ $= \frac{3}{\sqrt{2}} \times 0.05$ $= \frac{3}{20\sqrt{2}} \text{ or } \frac{3\sqrt{2}}{40} \text{ or } 0.106 \text{ units/s (3 s.f.)}$	3	M1 M1 A1	Differentiate r w.r.t. x Correct connected rates of change equation
10(c)	$\text{Area of triangle } OPR = \frac{1}{2}(1)(x) = \frac{x}{2}$ $\frac{dA}{dx} = \frac{1}{2}$		M1	Area of triangle

	$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$ $= \frac{1}{2} \times x^2$ $= \frac{x^2}{2}$ $\frac{x^2}{2} \geq 200$ $x^2 \geq 400$ $(x-20)(x+20) \geq 0$ $x \leq -20 \text{ or } x \geq 20$ Since $x > 0$, $x \geq 20$	4	 M1 M1 A1	Forming $\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$ Solving quadratic inequality
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Question	Answer	Marks	Partial Marks	Guidance
11(a)	$v = k \cos\left(t - \frac{\pi}{6}\right)$ When $t = 0$, $v = \sqrt{3}$ $\sqrt{3} = k \cos\left(-\frac{\pi}{6}\right)$ $k = 2$	2	M1 A1	Subst. $t = 0$ and v
11(b)	$v = 2 \cos\left(t - \frac{\pi}{6}\right)$ $2 \cos\left(t - \frac{\pi}{6}\right) = 0$ $t - \frac{\pi}{6} = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$ $t = \frac{2\pi}{3} \text{ or } \frac{5\pi}{3}$	2	B1, B1	
11(c)	$a = \frac{dv}{dt} = -2 \sin\left(t - \frac{\pi}{6}\right)$ At $t = 3$, $a = -2 \sin\left(3 - \frac{\pi}{6}\right) = -1.23 \text{ m/s}^2$	2	M1 A1	Finding $\frac{dv}{dt}$

11(d)	$s = \int 2 \cos\left(t - \frac{\pi}{6}\right) dt$ $= 2 \sin\left(t - \frac{\pi}{6}\right) + c$ When $t = 0, s = 0$. $0 = 2 \sin\left(0 - \frac{\pi}{6}\right) + c$ $c = 1$ $s = 2 \sin\left(t - \frac{\pi}{6}\right) + 1$ When $t = \frac{2\pi}{3}, s = 3$ When $t = \frac{5\pi}{3}, s = -1$ Distance travelled = $3 + (3 - (-1)) = 7$ m	5	M1 M1 M1 M1 A1	Finding s Find c or finding absolute value of definite integral between p and q. displacement at $t = p$ displacement at $t = q$
11(e)	It represents the displacement of the particle at $t = q$.	1	B1	Or equivalent